



Transportation Revenue and Expenditure analysis

Project Overview

This project presents an end-to-end exploratory data analysis (EDA) and business intelligence dashboard for Transportation funding analysis Dataset. The primary objective of this dataset is to analyze and compare transportation-related public finance statistics across U.S. states over multiple years and derive actionable insights to support data-driven transportation decision-making.

The project combines Python (Google Colab) for data preprocessing and exploratory analysis with Power BI for developing an interactive executive dashboard.

Project Objectives

- ◆ Clean and preprocess the raw transportation funding dataset.
- ◆ Perform Exploratory Data Analysis (EDA) to identify patterns, trends, and outliers.
- ◆ Analyze transportation statics by :
 - ◇ State
 - ◇ Mode of transportation
 - ◇ Year
- ◆ Study statistical properties such as skewness, kurtosis, and correlations.

Tools & Technologies

- ◆ Python
- ◆ Google Colab
- ◆ Pandas
- ◆ NumPy

- ◆ The combined population represented in the dataset is 4.04K crore.
- ◆ The analysis covers 52 states/territories, providing nationwide coverage.
- ◆ The average transportation value per population is 56.21, indicating the average financial allocation relative to population.

2. Revenue vs Expenditure

- ◆ Revenue contributes 76.38% (218.38K crore) of the total transportation finance.
- ◆ Expenditure accounts for 23.62% (67.53K crore).
- ◆ This indicates that transportation agencies generated substantially more revenue than they spent during the analysis period.

3. Transportation Mode Analysis

- ◆ Highways generate the highest transportation revenue among all transport modes.
- ◆ Transit has the highest population coverage, indicating that it serves the largest number of people.
- ◆ Air transportation contributes the smallest share of both revenue and population.

4. State-wise Performance

- ◆ Arizona records one of the highest transportation revenues.
- ◆ Tennessee and Georgia also demonstrate strong transportation financial performance.
- ◆ Wyoming shows comparatively higher expenditure than many other states.
- ◆ Financial performance differs significantly across states, reflecting variations in infrastructure investment and revenue sources.

5. Population vs Financial Value

- ◆ The scatter plot indicates a positive relationship between population and transportation financial value.
- ◆ States with larger populations generally generate higher transportation revenues.

◆ However, some states achieve relatively high financial values despite smaller populations, suggesting efficient revenue generation.

6. Year-wise Trend

◆ The average transportation financial value remained relatively stable over the three-year period:

- ◇ 2020: 31.02
- ◇ 2021: 30.93 (slight decline)
- ◇ 2022: 31.31 (highest average)

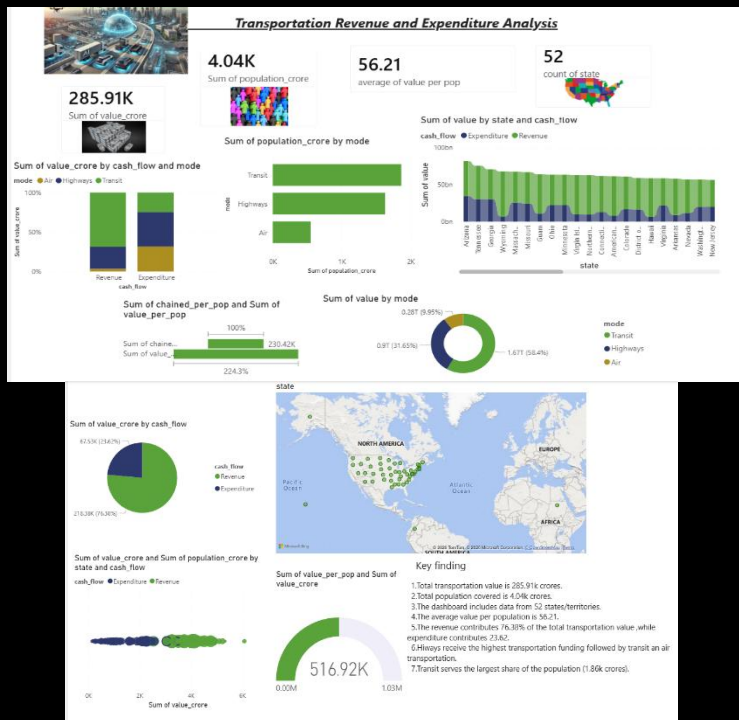
◆ The increase in 2022 suggests a recovery in transportation funding and financial activity.

Visualizations Included

- ◆ Bar Charts
- ◆ Line Charts
- ◆ Pie Charts
- ◆ Count Plots
- ◆ Correlation Heatmaps
- ◆ Scatter plots
- ◆ Violin plots

Power BI dashboard

- 1.Total transportation value is 285.91k crores.
- 2.Total population covered is 4.04k crores.
- 3.The dashboard includes data from 52 sates/territories
- 4.The average value per population is 56.21.
- 5.Revenue contributes 76.38% of the total transportation value, while expenditure contributes 23.62%
- 6.Hiways receive the highest transportation funding followed by transit an air transportation.
- 7.Transit serves the largest share of the population (1.86k crores)



🚀 Recommendation

Transportation revenue is considerably higher than expenditure, with significant differences across states. Future planning should emphasize efficient revenue generation, optimized spending, equitable funding based on population, and inflation-adjusted financial analysis. Benchmarking high-performing states and improving data quality will enable more informed budgeting and better transportation policy decisions.

Conclusion

The dashboard demonstrates that's the transportation sector is financially stable, with strong revenue generation, controlled expenditure and positive growth trends.

